

**BEFORE YOU READ — ANSWER FROM MEMORY**

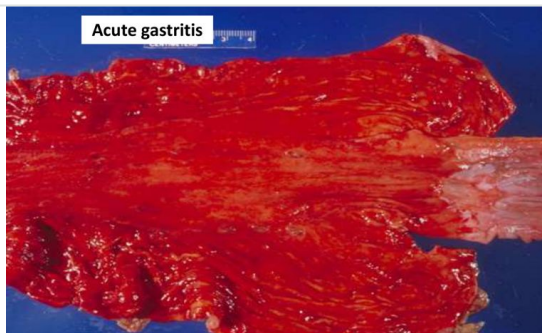
1. What is the exact depth that separates an **erosion** from an **ulcer**?
2. Which chronic gastritis causes **pernicious anemia**, and why does the other one not?
3. Are **duodenal or gastric** ulcers more common, and by how much?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

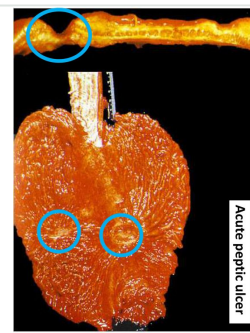
1 Acute disease — depth is everything

	Acute gastritis	Acute peptic ulceration
Depth of loss	Transient mucosal inflammation; surface epithelium intact . Erosion = loss of superficial epithelium above the muscularis mucosa	Tissue loss extends below the muscularis mucosa
Findings	Edema and congestion of mucosa; in severe cases erosion, ulceration, hemorrhage	Rounded, usually <1 cm ; single or (more often) multiple
Symptoms	May be asymptomatic, or epigastric pain, nausea, vomiting; severe: hematemesis, melena, rarely massive blood loss	Heals with complete re-epithelialisation days to weeks after the injurious factor is removed

Acute ulcer	Setting
Stress ulcer	Shock, sepsis or severe trauma
Curling ulcer	Severe burns
Cushing ulcer	Intracranial disease
Also	Complication of NSAID therapy



Acute gastritis: erythema, focal erosions



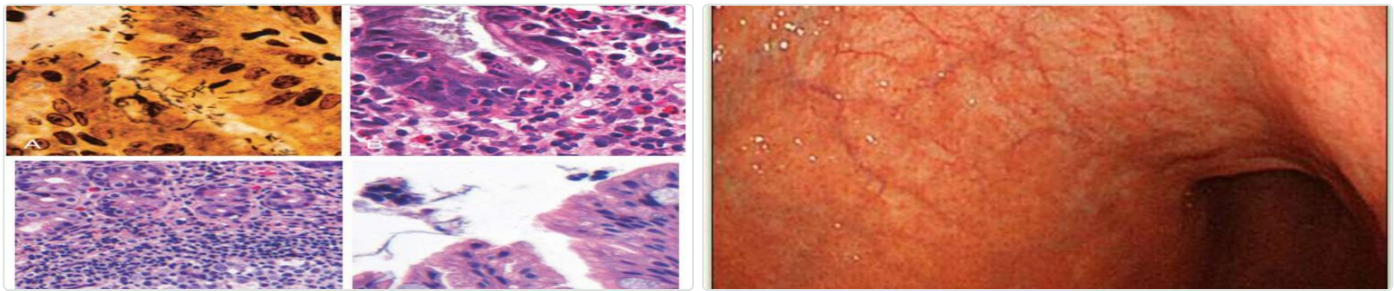
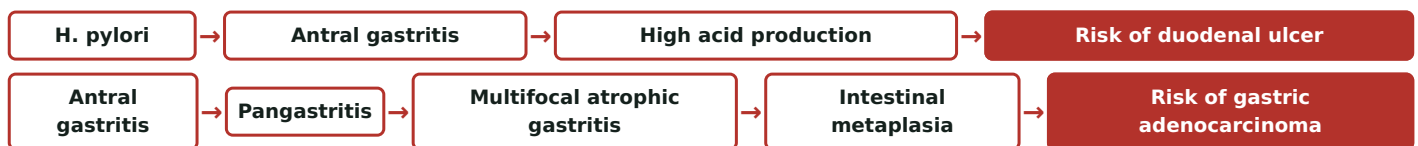
Acute peptic ulcer: hemorrhagic gastritis, focal ulcers (circled)

2 Chronic gastritis — H. pylori vs autoimmune

Less severe but more persistent symptoms than acute. Three types: **H. pylori** (the commonest), **autoimmune**, and **chemical (reactive)**.

	H. pylori gastritis	Autoimmune gastritis
Frequency	Commonest form of chronic gastritis	<10% of chronic gastritis, but the commonest cause of gastric atrophy and the commonest form in patients without H. pylori
Mechanism	Bacterial infection	Antibodies against parietal cells → loss of the cells that secrete HCl and intrinsic factor
Acid	Can cause hypochlorhydria , not achlorhydria	Severe parietal damage → achlorhydria
B12	Not associated with pernicious anemia (parietal damage less severe)	No intrinsic factor → ileal B12 absorption fails → megaloblastic (pernicious) anemia
Histology	Spiral bacilli (Warthin-Starry silver stain); neutrophils in epithelium and lamina propria; lymphoid aggregates with germinal centers; intestinal metaplasia	Gastric atrophy: whitish discoloration with visible vessels at endoscopy
Leads to	Chronic gastritis · PUD (almost all DU, majority of GU) · gastric adenocarcinoma · gastric MALToma (NHL)	Gastric atrophy · pernicious anemia

What H. pylori does — two routes



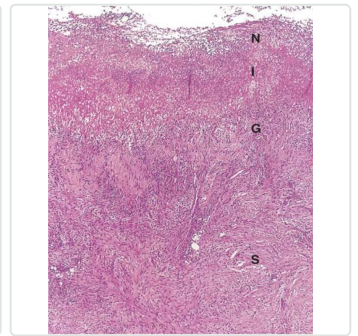
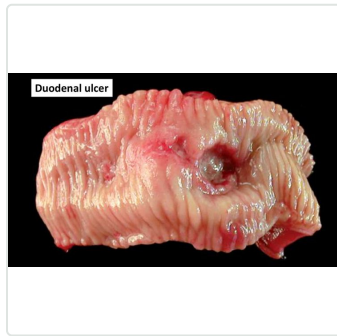
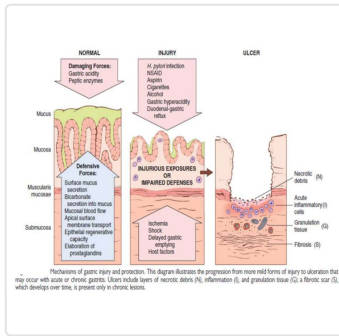
H. pylori silver stain · neutrophils · lymphoid aggregates · intestinal metaplasia

Gastric atrophy: whitish mucosa, visible vessels

3 Peptic ulcer disease (PUD)

Definition	Chronic lesions anywhere in the GIT exposed to acidic peptic juices.
Sites	Stomach (GU) · duodenum (DU) · lower (Barrett) esophagus · Meckel diverticulum with gastric heterotopia.
Pathogenesis	The same imbalance of mucosal defenses vs damaging forces that causes chronic gastritis: H. pylori , aspirin and NSAIDs, cigarette smoking, increased gastric HCl and pepsin.
Morphology	Solitary in >80% . Round to oval, sharply punched-out ; regular margins with radiating mucosal folds; smooth, clean base (peptic digestion of the exudate).

	Duodenal ulcer (DU)	Gastric ulcer (GU)
Frequency	Four times more common	Less common
Site	Within a few cm of the pyloric valve, anterior duodenal wall	Near the interface of body and antrum, lesser curvature
H. pylori	Present in almost all patients	Present in the majority
Malignant change	Less likely	More likely (still very rare)



Damaging forces vs defenses: normal
 → injury → ulcer

Gastric ulcer: sharp edge, radiating folds

Duodenal ulcer: punched-out, clean base

Layers: N necrotic, I inflammatory, G granulation, S scar

Histology

Four layers from the lumen down: **Necrotic** → **Inflammatory** → **Granulation tissue** → **Scar** (fibrosis).

Complications

Hemorrhage (commonest; occult or massive) · pyloric (gastric outlet) obstruction · perforation and penetration · combined GU and DU · malignant transformation of a benign ulcer (very rare; GU more than DU).

P Must-remember pearls

- **Erosion stops above the muscularis mucosa; ulceration goes below it.**
- **Curling = burns; Cushing = intracranial disease;** stress ulcers follow shock, sepsis or severe trauma.
- **Only autoimmune gastritis causes achlorhydria and pernicious anemia;** *H. pylori* does not, because parietal damage is less severe.
- **DU are four times commoner than GU,** on the anterior duodenal wall, with *H. pylori* in almost all; GU sit on the lesser curvature and are the ones that (rarely) turn malignant.
- **Hemorrhage is the commonest PUD complication.**

Q Test yourself

1. A patient with extensive burns develops upper GI bleeding from multiple small acute gastric ulcers. Which eponym applies?

- A. Curling ulcer
- B. Cushing ulcer
- C. Meckel ulcer
- D. Zollinger ulcer
- E. Marjolin ulcer

2. A 60-year-old has megaloblastic anemia and achlorhydria. Gastric biopsy shows atrophy without organisms. Which mechanism explains her anemia?

- A. *H. pylori*-induced hypochlorhydria
- B. Chronic NSAID use
- C. Chemical (reactive) gastritis
- D. Antibodies against parietal cells with loss of intrinsic factor
- E. Zinc deficiency

3. A gastric biopsy shows tissue loss extending below the muscularis mucosa. Which term applies?

- A. Erosion
- B. Intestinal metaplasia
- C. Ulceration
- D. Atrophy
- E. Hyperplasia

4. Which statement about peptic ulcers is correct?

- A. Gastric ulcers are four times commoner than duodenal ulcers
- B. Duodenal ulcers usually arise on the posterior wall
- C. Malignant transformation is commoner in DU than GU
- D. *H. pylori* is present in only a minority of DU
- E. Duodenal ulcers are four times commoner than gastric ulcers

5. H. pylori antral gastritis leads to high acid production. Which consequence follows most directly?

- A.** Pernicious anemia
- B.** Increased risk of duodenal ulcer
- C.** Achlorhydria
- D.** Loss of intrinsic factor
- E.** Linitis plastica

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — Curling ulcers follow severe burns. Cushing ulcers accompany intracranial disease; stress ulcers follow shock, sepsis or severe trauma.
- 2 • D** — Autoimmune gastritis destroys parietal cells, removing HCl and intrinsic factor so ileal B12 absorption fails: pernicious anemia.
- 3 • C** — By definition ulceration extends below the muscularis mucosa; erosion is loss of superficial epithelium above it.
- 4 • E** — DU are four times commoner, usually on the anterior duodenal wall; malignant change favours GU; H. pylori is in almost all DU.
- 5 • B** — Per the slide diagram, antral gastritis → high acid production → increased risk of DU; the other route leads through atrophy and metaplasia to adenocarcinoma.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL1, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.